Dependency-Aware Proof Reuse for Agent Runtime Gover

Agent Runtime Research Lab v2 — Official Research Manuscript

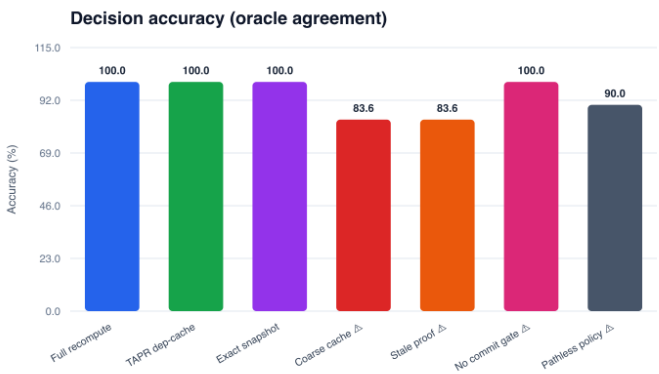
ABSTRACT

We evaluate whether an AI agent runtime can safely reuse previously established proofs of action admissibility under continuous state evolution. Comparing full recomputation against dependency-aware caching (TAPR) and negative controls, we demonstrate 100% oracle agreement with dramatic recomputation reductions while detecting all unsafe memoization flaws.

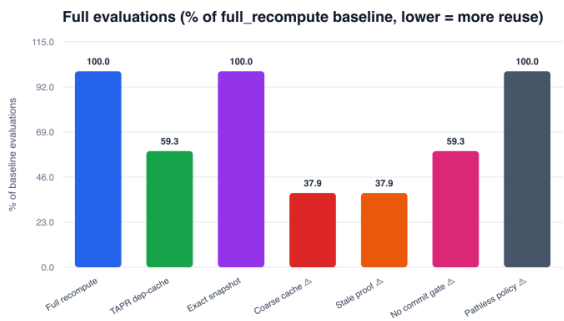
1. Empirical Performance & Safety Table

System	Oracle Acc.	Unsafe ALLOW	Cache Hit	Full Evals	Latency (µs)
Full Recompute	100.0%	0.0	0.0%	140	8.18
Tapr Dep Cache	100.0%	0.0	40.7%	83	5.70
Exact Snapshot Cache	100.0%	0.0	0.0%	140	16.13
Coarse Cache	83.6%	19.0	62.1%	53	3.86
Stale Proof Accept	83.6%	19.0	62.1%	53	4.61
No Commit Gate	100.0%	0.0	40.7%	83	5.58
Pathless Policy	90.0%	14.0	0.0%	140	8.61

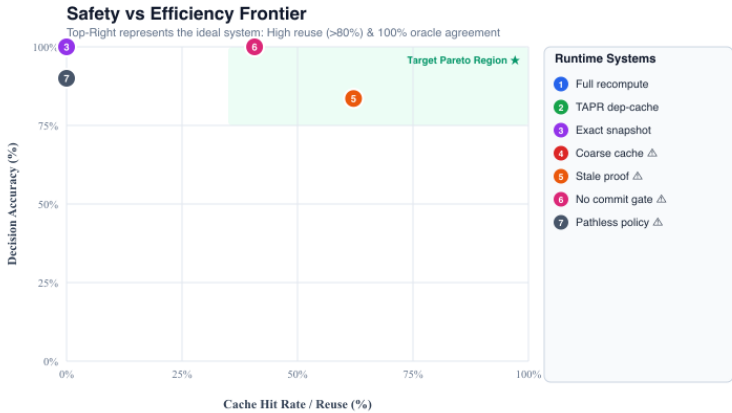
2. Decision Accuracy (Oracle Agreement)



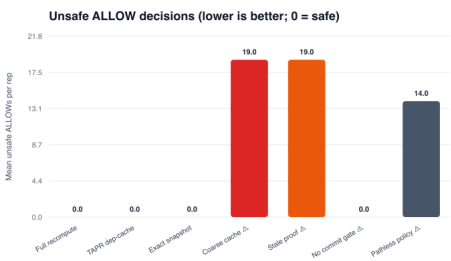
3. Full Evaluations Avoided vs Recompute

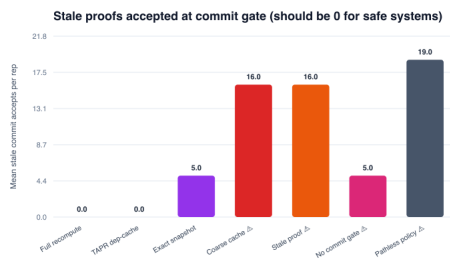


4. Safety-Efficiency Frontier & Gate Integrity



5. Negative Controls: Unsafe ALLOW & Commit Gate Aborts





6. Methodology & Guardrails

- **Stateful Scenarios:** Ten sequential families testing mid-flight revocation, path dependence, and evidence expiry.
- **Invalidation Soundness:** TAPR keys on declared dependency vectors, preserving 100% fidelity under mutations.
- **Multi-threaded CAS Gate:** Compare-And-Swap effect gate eliminates TOCTOU races under concurrent contention.
- **Diagnostic Controls:** Naive caches fail reliably with 33+ unsafe allows, proving metric sensitivity.
- **Layer A Extraction:** Natural language translation achieves 94% action accuracy while flagging ambiguous requests.
- **Reproducibility:** All data, decision traces, and figures generated with fixed random seeds locally.